Sinan Emre

Researcher in Mechanical Design of Legged Robots and Actuators

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PhD researcher, IIT Dynamic Legged Systems lab. Available for postdoctoral positions from February 2027.

Research interests

Robot design and actuation: legged robot mechanisms, variable transmissions; compact electric, series elastic and hydraulic actuation.

Methods: topology and design optimization, FEM, simulation and experimental validation on quadrupeds.

Education

PhD, Bioengineering and Robotics Nov 2023 – present
Istituto Italiano di Tecnologia / University of Genova, Italy. Dynamic Legged Systems lab.
Thesis (working title): *Design of the Next Generation Quadruped Robots.*

MSc, Mechanical Engineering Sep 2019 – Aug 2022
Özyeğin University, Istanbul, Turkey. 100% merit scholarship, Özyeğin merit grant, TÜBİTAK grant.
Thesis: *Design, development, and real-life implementation of a high power quadruped robot.* Sole designer of the robot and its actuators; two prototypes built and tested in gait experiments.
Supervisors: Ramazan Ünal, Barkan Uğurlu.

BSc, Mechanical Engineering Sep 2013 – Aug 2018
Özyeğin University, Istanbul, Turkey. 50% merit scholarship.

Research experience

Istituto Italiano di Tecnologia (IIT), Dynamic Legged Systems lab, Genova, Italy Feb 2023 – present

PhD Researcher in Bioengineering and Robotics Nov 2023 – present

Research Fellow in Mechanical Engineering Feb 2023 – Nov 2023

- **Carpal-claw mechanism:** carried out the mechanical design of a mechanism for high-obstacle negotiation by quadruped robots, and tested it on Unitree Aliengo [C3].
- **Load-based variable transmission:** passive knee mechanism (pre-tensioned torsional springs and a four-bar linkage) that increases the knee transmission ratio under load, by up to 40% in simulation; built a 3D-printed prototype [C2].
- **Compact hydraulic power unit:** simulated and experimentally evaluated the unit, and integrated it on the HyQReal2 hydraulic quadruped [C1].
- **Stair-climbing study:** built the simulations for a study of the optimal body height for quadrupeds walking on stairs [C4].
- **HyQReal and Unitree robots:** system integration, experiments, mechanical maintenance and public demos on HyQReal, including work in the INAIL project; experiments on Unitree Aliengo; demos on Unitree Go2.

Özyeğin University, Biomechatronics Lab, Istanbul, Turkey Jun 2018 – Nov 2022

Research and Design Engineer Apr 2021 – Nov 2022

- Developed an additively manufactured actuator unit and quadruped prototype using composite materials; built FEM models of composite structures and of infill types under load; carried out quality control and experiments on actuator units and quadruped prototypes.

Research Engineer, CAD Design and Optimization Apr 2020 – Apr 2021

- Sole designer of a compact actuator unit for quadrupeds, 48 Nm continuous torque at 1.6–1.7 kg [C5], and of a topology-optimized 12-DOF quadruped prototype driven by these actuators.
- Built parametric FEM models in Ansys, selected materials and ran topology optimization for critical load cases; built dynamic simulation models of floating-base robots; handled manufacturing follow-up and quality control.

Lab and Teaching Assistant; earlier Lab Assistant, Mechanical Design Jun 2018 – Apr 2020

- CoEx lower-body exoskeleton (TÜBİTAK project), from its start: designed the limbs, verified the series elastic actuator, ran FEM on mechanical parts and tracked manufacturing; designed mechanism parts for pneumatic muscles [J1, C6, P1].
- Topology optimization of the torsional spring in a series elastic actuator, with design verification [C7].

Publications

Journal article

- [J1] M. C. Yıldırım, P. Şendur, A. T. Kansızoğlu, U. Uras, O. Bilgin, **S. Emre**, G. G. Yapıcı, M. Arık, B. Uğurlu. Design and development of a durable series elastic actuator with an optimized spring topology. *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*, 235(24), pp. 7848–7858, 2021. doi.org/10.1177/09544062211020337

Conference papers

- [C1] **S. Emre**, M. Villa, V. Barasuol, C. Semini. Design and performance evaluation of a compact hydraulic power unit for legged robots. *IEEE/ASME AIM*, 2026.
- [C2] **S. Emre**, V. Barasuol, M. Villa, C. Semini. Load-based variable transmission mechanism for robotic applications. *IEEE ICAR*, pp. 73–79, 2025. Preprint: arxiv.org/abs/2512.15448
- [C3] V. Barasuol, **S. Emre**, V. S. Medeiros, A. Bratta, C. Semini. Introducing the carpal-claw: a mechanism to enhance high-obstacle negotiation for quadruped robots. *IEEE ICRA*, pp. 3457–3463, 2024.
- [C4] V. Barasuol, **S. Emre**, C. Semini. Stair-climbing charts: on the optimal body height for quadruped robots to walk on stairs. *CLAWAR*, Springer, pp. 251–262, 2023.
- [C5] **S. Emre**, B. Akın, E. C. Özçınar, B. Balcı, P. Şendur, Ö. Bebek, R. Ünal, B. Uğurlu. Topology optimization-based design and development of a compact actuator with a high torque-to-weight ratio for quadrupeds. *ACTUATOR 2022*, VDE, pp. 1–4, 2022.
- [C6] M. C. Yıldırım, A. T. Kansızoğlu, **S. Emre**, M. Derman, S. Çoruk, A. F. Soliman, P. Şendur, B. Uğurlu. Co-Ex: A torque-controllable lower body exoskeleton for dependable human-robot co-existence. *IEEE ICORR*, pp. 605–610, 2019. doi.org/10.1109/ICORR.2019.8779407
- [C7] Yüksek tork çıkışlı seri elastik eyleyiciler için burulma yayı tasarımı ve topoloji eniyileştirmesi (Torsion spring design and topology optimization for high-torque series elastic actuators). *TORK*, 2019. In Turkish.

Patent

- [P1] M. C. Yıldırım, B. Uğurlu, P. Şendur, **S. Emre**, M. Derman, S. Çoruk. Wearable lower extremity exoskeleton. *US Patent 12,521,298 B2*, 2026; *EP 3 986 353 B1*, 2024.

Teaching

Lab and Teaching Assistant, Özyeğin University (2019–2020): lab sessions for CS-100; exam sessions for CS-100 and mathematics courses.

Skills

Design and manufacturing: Siemens NX, CATIA, SolidWorks, Autodesk Inventor; engineering drawings with GD&T and tolerances; additive manufacturing and composite materials; work with machine shops and suppliers.

Analysis and experiments: FEM (Ansys, Abaqus, ANSA-META), topology optimization, MSC Adams, MATLAB, NI LabVIEW, LaTeX.

Robot software: ROS, ROS 2.

Languages and interests

Languages: Turkish (native) | English (fluent) | German (A2–B1) | Italian (A2)
Interests: Basketball, professional licensed player (2011–2013). Sailing, 1-star licence (2017). Wing Chun kung fu, self-defence instructor since 2022.